SS - IMP. PYQ Analysis

SYSTEM SOFTWARE | BE05000261 | Sem 5 | GTU

New Syllabus 2024-25 | P. Analyzed: 13 (Summer/Winter 2018 – 2025)

Chapter Reference

Ch. 1 - Overview of System Software: Introduction, Life Cycle of Source Program, Levels of System Software

Ch. 2 - Overview of Language Processors: Language Processing Activities, Symbol Tables

Ch. 3 - Assemblers: Design, Types, Two-Pass/One-Pass, Advanced Assembly Process

Ch. 4 - Macro and Macro Processors: Definition, Expansion, Nested Calls, Macro Preprocessor Design

Ch. 5 - Linkers and Loaders: Relocation, Linking, Loading Schemes

Ch. 6 - Scanning and Parsing: Grammars, Top-Down/Bottom-Up Parsing, LEX, YACC

Ch. 7 - Compilers: Semantic Gap, Binding, Code Optimization

Ch. 8 - Interpreters & Debuggers: JVM, Types of Errors, Classification of Debuggers

> 📌 **Ch. 9** - Programming Languages: High-Level Language features, Data Types, Storage Allocation, Scope, Extensibility *(Zero PYQ - Blind Spot, entirely new unit)*

M.IMP Questions - High → Low Priority

📌 **Priority Rule:** Freq ≥ 5/13, OR a repeating numerical/diagram type, AND confirmed present in new syllabus

1. Symbol Table + Intermediate Code Generation from Assembly Program *(EQU / ORIGIN / LTORG, forward reference handling)*

(Ch. 3) | Freq: **8x** | Max Marks: **7** | Num: 8

2. Remove Left Recursion from Given Grammar

(Ch. 6) | Freq: **9x** | Max Marks: **4** | Num: 9

3. OPTAB, SYMTAB, LITTAB, POOLTAB - Describe Data Structures

(Ch. 3) | Freq: **6x** | Max Marks: **4–7** | Theory: 6

4. Construct LL(1) Parsing Table (+ Parse a String Using It)

(Ch. 6) | Freq: **7x** | Max Marks: **4–7** | Num: 7

5. Macro Tables: KPDTAB, MDT, EVTAB, SSTAB / MNT - Use and Fields

(Ch. 4) | Freq: **5x** | Max Marks: **4–7** | Theory: 5

6. Compare Macros and Subroutines *(code space, execution speed, processing, flexibility)*

(Ch. 4) | Freq: **5x** | Max Marks: **3–7** | Theory: 5

7. AIF / AGO / ANOP - Expansion Time Statements with Expansion Time Loop

(Ch. 4) | Freq: **5x** | Max Marks: **3–7** | Theory: 5

8. Write a Macro Definition for a Given Computation Using Parameters

(Ch. 4) | Freq: **5x** | Max Marks: **3–7** | Num: 5

9. Positional vs Keyword vs Default-Value Parameters in Macros

(Ch. 4) | Freq: **5x** | Max Marks: **3–4** | Theory: 5

10. Nested Macro Call with Example

(Ch. 4) | Freq: **5x** | Max Marks: **3–4** | Theory: 5

11. Self-Relocating Program - Explain / Characteristics

(Ch. 5) | Freq: **6x** | Max Marks: **3–7** | Theory: 6

12. Overlay Structured Program - Definition + Execution / Linking

(Ch. 5) | Freq: **6x** | Max Marks: **3–7** | Theory: 6

13. Compile-and-Go Loader - Explain with Example

(Ch. 5) | Freq: **5x** | Max Marks: **3–4** | Theory: 5

14. Program Relocation - What It Is / How Performed by Linker

(Ch. 5) | Freq: **6x** | Max Marks: **3–7** | Theory: 6

15. Absolute Loader - Explain with Example

(Ch. 5) | Freq: **5x** | Max Marks: **3–4** | Theory: 5

16. Pure vs Impure Interpreter - Differentiate / Explain

(Ch. 8) | Freq: **6x** | Max Marks: **3–7** | Theory: 6

17. JVM - Explain / Describe with Diagram

(Ch. 8) | Freq: **4x** | Max Marks: **4–7** | Theory: 3 Diag: 1

18. Code Optimization Techniques - List / Explain with Example

(Ch. 7) | Freq: **6x** | Max Marks: **4–7** | Theory: 6

19. Triples, Quadruples, Indirect Triples - Generate for a Given Expression

(Ch. 7) | Freq: **6x** | Max Marks: **4–7** | Num: 6

20. Binding and Binding Times - Explain

(Ch. 7) | Freq: **4x** | Max Marks: **3–4** | Theory: 4

21. One-Pass vs Two-Pass Assembler - Differentiate / Forward Reference Handling

(Ch. 3) | Freq: **5x** | Max Marks: **3–7** | Theory: 5

22. Types of Grammar - Explain / List

(Ch. 6) | Freq: **5x** | Max Marks: **3** | Theory: 5

23. Ambiguous Grammar / Ambiguity in Grammatic Specification

(Ch. 6) | Freq: **5x** | Max Marks: **3** | Theory: 5

24. Linker vs Loader - Differentiate / Compare

(Ch. 5) | Freq: **5x** | Max Marks: **3** | Theory: 5

🟡 IMP Questions - High → Low Priority

📌 **Priority Rule:** Freq 2-4/13 present in new syllabus, OR low frequency but 07-mark questions appearing 2x+

1. First and Follow Set + Check if Grammar Is LL(1)

(Ch. 6) | Freq: **3x** | Max Marks: **7** | Num: 3

2. Shift-Reduce / Bottom-Up Parsing Using Handle and Simple Phrase

(Ch. 6) | Freq: **3x** | Max Marks: **7** | Num: 3

3. Design of Macro Preprocessor - Schematic + Task Explanation

(Ch. 4) | Freq: **4x** | Max Marks: **3–7** | Theory: 4

4. Flowchart: One-Pass Macro Processor / TII Maintenance

(Ch. 4) | Freq: **2x** | Max Marks: **3–7** | Diag: 2

5. REPT and IRP Statements - Explain with Example

(Ch. 4) | Freq: **3x** | Max Marks: **4** | Theory: 3

6. Lexical vs Semantic Macro Expansion

(Ch. 4) | Freq: **3x** | Max Marks: **3** | Theory: 3

7. Advanced Assembler Directives - Explain with Example

(Ch. 3) | Freq: **3x** | Max Marks: **7** | Theory: 3

8. Forward Reference - Define + Solve Using Back-Patching

(Ch. 3) | Freq: **3x** | Max Marks: **3–7** | Theory: 3

9. PASS-1 and PASS-2 Assembler Tasks - Explain

(Ch. 3) | Freq: **2x** | Max Marks: **7** | Theory: 2

10. Bootstrap Loader - Explain

(Ch. 5) | Freq: **2x** | Max Marks: **3** | Theory: 2

11. Relocating Loader vs Absolute Loader / Direct Linking Loader

(Ch. 5) | Freq: **2x** | Max Marks: **4** | Theory: 2

12. Linker Resolving External References - Explain in Detail

(Ch. 5) | Freq: **2x** | Max Marks: **7** | Theory: 2

13. Object Module: Address / Size / Load-Origin / Relocation-Factor Computation

(Ch. 5) | Freq: **2x** | Max Marks: **4** | Num: 2

14. Recursive Descent Parsing Algorithm - Explain / Trace

(Ch. 6) | Freq: **3x** | Max Marks: **7** | Theory: 2 Num: 1

15. Left Factoring - Explain with Example / Need for It

(Ch. 6) | Freq: **2x** | Max Marks: **3–4** | Theory: 2

16. Operator Precedence Grammar - Convert + Build Matrix + Parse String

(Ch. 6) | Freq: **3x** | Max Marks: **7** | Num: 3

17. Top-Down vs Bottom-Up Parser - Compare

(Ch. 6) | Freq: **2x** | Max Marks: **3** | Theory: 2

18. DAG (Directed Acyclic Graph) - Explain with Example

(Ch. 7) | Freq: **2x** | Max Marks: **3** | Theory: 2

19. Activation Records / Stack-Based Memory Allocation - Parameters and Limitations

(Ch. 7) | Freq: **2x** | Max Marks: **3** | Theory: 2

20. Static vs Dynamic Binding

(Ch. 7) | Freq: **1x** | Max Marks: **4** | Theory: 1

21. Classification of Debuggers

(Ch. 8) | Freq: **3x** | Max Marks: **4** | Theory: 3

22. Compiler vs Interpreter - Differentiate

(Ch. 8) | Freq: **3x** | Max Marks: **3** | Theory: 3

23. Life Cycle of Source Program - Diagram + Explain

(Ch. 1) | Freq: **3x** | Max Marks: **7** | Theory: 2 Diag: 1

24. User-Centric vs System-Centric View of System Software

(Ch. 1) | Freq: **2x** | Max Marks: **3–4** | Theory: 2

25. DFA Construction / Minimization for Given Regular Expression

(Ch. 6) | Freq: **2x** | Max Marks: **7** | Num: 2

26. Definitions Cluster *(Semantic Gap, Specification Gap, Execution Gap, Language Migrator, Token, Handle, etc.)*

(Ch. 1 / 2) | Freq: **6x** | Max Marks: **4** | Def: 6

🔵 New Syllabus Blind Spots - High → Low Priority

📌 **Zero PYQ Coverage** - Ch. 9 is entirely new. Predictions below are based on general GTU paper-setting patterns for this topic type, not on any past paper.

1. Features / Importance of High-Level Languages vs Low-Level

(Ch. 9) | Likely: **Theory** | Max Marks: **3** *(Predict)*

2. Data Types and Data Structures in Programming Languages - Explain

(Ch. 9) | Likely: **Theory** | Max Marks: **4** *(Predict)*

3. Storage Allocation Techniques (Static, Stack, Heap) and Scope of Names

(Ch. 9) | Likely: **Theory** | Max Marks: **7** *(Predict)*

4. Accessing Flexibility / Functional Modularity - Explain

(Ch. 9) | Likely: **Theory** | Max Marks: **3–4** *(Predict)*

5. Asynchronous Operation and Extensibility in Programming Languages

(Ch. 9) | Likely: **Theory** | Max Marks: **3** *(Predict)*

6. Compile Time vs Run Time Considerations in Language Design

(Ch. 9) | Likely: **Theory** | Max Marks: **4** *(Predict)*

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